

LEC: A/B Testing and Launch Recommendation:

Agenda:

① Business sense towards A/B Testing

② How to launch A/B Testing

↳ Developing the Hypothesis

↳ Design A/B Tests

↳ Evaluate A/B Tests

↳ Making decisions.

③ Launch Recommendations A A/B Testing setup pitfalls.

Examples of A/B Testing:

- ① Multivariate Testing :
- ② Split Testing : (version A vs version B)
- ③ Conversion Rate Optimization :
- ④ Landing Page Optimization :

⑤ Online Experimentation :

Judgement Criteria A General Framework:

① Design an Experiment

① structure :

② Right Flow of steps :

③ Develop the right Hypothesis :

→ ④ Sample size calculation :

→ ⑤ Set the duration

⑦ Pitfalls : Common mistakes

② Focus on key areas :-

① Choose the right metric : NIM

② Segment your sample

③ Making decisions from Results.

Case Study:

Problem Stmt:

How would you determine if adding
colored backgrounds to Facebook statuses
increases the user engagement?



Post to Facebook

POST



Mark Wilson

 Friends ▼

 Miskowice ✕

What's on your mind?



1. Clarify the goal of the feature / Idea conception.

↳

↳

↳

↳

2. Discuss the Metrics Impacted by the Feature and the available data.

↳ % User Engagement - NSM

↳ DAU - Daily Active Users

↳ Guardrail metrics:

↳

↳

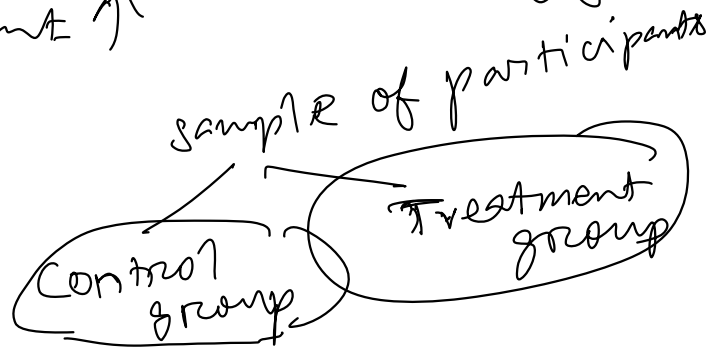
3. How to Design an Experiment?

① Setup of Hypotheses

↳ H_0 : The feature does not make a difference on user engagement

↳ H_a : User engagement $\uparrow$

② Z proportion test.



③ select participants (Control vs Treatment)

④ Calculate the sample size:

✓ Baseline Engagement Rate (BER):

Current user engagement is 45%.

✓ MDE (Minimum Detectable Effect):

Business goal (46%)

$$\text{MDE} = (46 - 45)\% = 1\%$$

✓ significance level: 95%

✓ power = 80%

↳ 40K users in each group $\begin{matrix} \nearrow C \\ \searrow T \end{matrix}$

Extra (out of scope):

		Z scores
95% confidence	$\alpha = 0.05$	$\rightarrow 1.96$
80% power	$\beta = 0.2$	$\rightarrow 0.84$

$$Z_{\alpha/2} = 1.96$$

$$Z_{\beta} = 0.84$$

$$n = \frac{(Z_{\alpha/2} + Z_{\beta})^2 * (p_1(1-p_1) + p_2(1-p_2))}{(p_2 - p_1)^2}$$

$p_1 = \underline{\text{baseline conversion}}$

$p_2 = \underline{\text{expected conversion}}$

$$p_2 - p_1 = \text{MDE}$$

$$n = 38,870 \approx 40k \text{ per group.}$$

⑥ Duration :

Assumptions

Traffic per day : 5000 daily users

$$\text{Required sample size} = 40k + 40k \\ = 80k$$

$$\text{Duration} = \frac{80k \text{ people}}{5k \text{ people / day}} = 16 \text{ days}$$

(f) Test and Monitor Metrics

(4) Avoid common pitfalls :-

(A) Experimental Design Bias:

✓ $\rightarrow$ Primacy Effect:

User engagement: Week 1: 45%, Week 2: 48%,
Week 3: 62%.

✓ $\rightarrow$ Novelty Effect:

User engagement: Week 1: 65%, Week 2: 68%,
Week 3: 52%.

③ Group Inferences :

① Network Effect :

② Outcome Bias :

⑤ Recommendations Based on
Experiment Results — To Launch or
not Launch.

⑥ Link results to the Goal
A Business Impact:

↳ 1% Increase in user engagement.

↳ ? → Business value

↳ Revenue Impact: 1% → \$20M

1% → \$50k

↳ Effort vs Return.

③ Handling conflicting results

④ short term vs Long term Impact,

Final Decision Making :

An e-commerce website wants to optimize its checkout page. They run an A/B test where half of the visitors see a green "Buy Now" button, while the other half sees a red "Buy Now" button. What could be the goal of this experiment? Choose the most appropriate answer.

30 users have participated



A

The goal is to determine which color increases the click-through rate and leads to more purchases.

77%

B

The goal is to assess whether changing the button color impacts customer satisfaction with the website design.

23%

C

The goal is to analyze the impact of button color on the average time spent on the checkout page.

0%

A media company is testing different subject lines for its weekly newsletter. They send two versions of the newsletter, each with a different subject line, to a subset of their subscribers. What would be the most appropriate action item for the media company as per the ongoing experiment?

25 users have participated

A

They conduct a survey to gather feedback from subscribers about the subject line for their respective group.

8%

B

They change the font style & color among the two versions of the newsletter to see if it has any impact on subscriber interaction.

20%

C

They increase the frequency of newsletter distribution for one group to see if more frequent newsletters result in higher engagement.

12%



D

They measure open rates to determine which subject line is more effective at capturing reader attention.

60%

An online retailer is experimenting with the layout of its product pages. They test two variations: one with the product description at the top and one with customer reviews at the top. What would be the most appropriate choice of next step in the currently ongoing experiment?

38 users have participated

A

They analyze the number of social media shares for each product to understand the impact of the layout variations on brand popularity.

8%

 B

The company measures the click-through rate and conversion rate to determine which layout leads to more sales.

58%

C

They conduct a survey to gather customer preferences on the more appealing layout of the product pages of the website.

34%

A streaming service wants to enhance its content recommendation algorithm. They split their user base into two groups: one group receives recommendations from the current algorithm, and the other group experiences recommendations from a new algorithm. What would be the most appropriate choice of next step in the currently ongoing experiment?

29 users have participated



- | | | |
|---|--|-----|
| A | They analyze the number of products recommended during the testing period to assess the impact of the algorithm changes. | 0% |
| B | They monitor the recommendations generated by the new algorithm and compare them with the recommendations from their rival platforms. | 3% |
| C | They compare user engagement metrics, such as time spent watching content and user ratings among the two algorithms to decide which one is to be kept as default for the entire user base. | 97% |
| D | They analyze the variety of categories of products recommended during the testing period to assess the impact of the algorithm changes. | 0% |